

Homework 1

PENG Qiheng

Student ID 225040065

February 28, 2026

Problem 1.

(a) **INTRO1: What does “act rationally” mean?**

In AI, “act rationally” means an agent chooses the action that maximizes the expected outcome (utility/performance) based on available information and constraints. It does **not** mean the agent is morally good or knows everything; it means decision-making is logically consistent with its objective.

Examples:

- A GPS navigation system selects the route with minimum expected travel time, even if it is not the shortest distance.
- In online ad bidding, a bidder sets bids to maximize expected conversion value under a budget, instead of bidding blindly.

Problem 2.

(a) **INTRO2: One task humans do not do well**

Humans are generally weak at consistently comparing large amounts of high-dimensional visual data. For example, matching one face against millions of surveillance images is slow and error-prone for humans, while computer vision systems can do this at large scale with stable accuracy.

Problem 3.

(a) **BFS1 (8-puzzle search count)**

From the slide, the goal node is reached in the level-5 expansion stage, and the marked index indicates it is the 27-th searched state. Therefore, the number of searched states is:

$$N_{\text{searched}} = 27.$$

Problem 4.

(a) **DFS1 (draw DFS tree and classify edges)**

Using the graph on page 4 and the visit order shown on the slide:

$$u \rightarrow v \rightarrow t \rightarrow s \rightarrow r \rightarrow w.$$

DFS tree edges are:

$$(u, v), (v, t), (t, s), (t, r), (r, w).$$

Non-tree edges are back edges to ancestors:

$$(u, s), (v, w).$$

This matches the slide note: backtrack from s , then w , then r , then t .

Problem 5.

(a) **DFS2 (discovery/finish times)**

According to the DFS recursion sequence in the figure, the entry/exit times are:

$$u : (1, 12), \quad v : (2, 11), \quad t : (3, 10), \quad s : (4, 5), \quad r : (6, 9), \quad w : (7, 8).$$

(Format: $d[v], f[v]$.)

Problem 6.

(a) **DFS3 (alphabetical visiting order)**

For the directed graph on page 6 (with vertices and adjacency lists both processed alphabetically), one valid DFS visit order is:

$$q, s, v, w, t, x, z, y, r, u.$$

A corresponding backtracking (finish) order is:

$$w, v, s, z, x, y, t, q, u, r.$$

Problem 7.

(a) **Dijkstra1**

Source is a . Final shortest distances are:

$$\begin{aligned} d(a) &= 0, \quad d(b) = 4, \quad d(h) = 8, \quad d(g) = 9, \quad d(f) = 11, \\ d(c) &= 12, \quad d(i) = 14, \quad d(d) = 19, \quad d(e) = 21. \end{aligned}$$

One shortest-path tree predecessor set is:

$$\begin{aligned} \pi(b) &= a, \quad \pi(h) = a, \quad \pi(g) = h, \quad \pi(f) = g, \\ \pi(c) &= b, \quad \pi(i) = c, \quad \pi(d) = c, \quad \pi(e) = f. \end{aligned}$$

Problem 8.

(a) **A*1**

Start node: A , goal node: T .

Main expansions (with $h(B) = 16$):

- Expand A : $g(A) = 0, f(A) = 30$; generate X with $g = 17, f = 44$.
- Expand X : generate B ($g = 33, f = 49$), C ($g = 28, f = 45$).
- Expand C : generate W ($g = 39, f = 48$), L ($g = 48, f = 81$).
- Expand W : generate T ($g = 48, f = 48$), K ($g = 49, f = 68$).
- Expand T (goal), stop.

Final path:

$$A \rightarrow X \rightarrow C \rightarrow W \rightarrow T,$$

with total cost 48.

Problem 9.

(a) **A*2 (change $h(B)$ from 16 to 6)**

With $h(B) = 6$, node B is expanded earlier, so the expansion order changes. However, the final optimal path is still:

$$A \rightarrow X \rightarrow C \rightarrow W \rightarrow T,$$

with the same optimal cost 48.

Conclusion: both settings are admissible, so optimality is preserved; the difference is mainly search efficiency/order (smaller $h(B)$ is less informative and may cause extra exploration).

Problem 10.

(a) **A*3 (optional, textbook p.114 Problem 3.7)**

- (a) For polygonal shortest path planning, a standard finite state set is the visibility-graph nodes: all obstacle vertices plus S and G . If obstacle vertices total is V , then number of states is $V + 2$.
- (b) A shortest collision-free path is piecewise linear and only changes direction at obstacle vertices; otherwise a bend in free space can be shortened by a straight segment (triangle inequality). Therefore shortest paths go along visibility edges and touch polygon boundaries at vertices.